

THANH NAM TRAN

Silver Spring, MD 20904

(301) 232-4963

ttran19@umbc.edu ▪ [linkedin.com/in/thanh-nam-tran/](https://www.linkedin.com/in/thanh-nam-tran/) ▪ github.com/ttran293

Technical Skills

Programming & OOP: Python, C++, Object-Oriented Programming (OOP), Design Patterns

Databases: Microsoft SQL Server, PostgreSQL, MongoDB, ORM (SQLAlchemy, Django ORM)

Full Stack Development: HTML, CSS, JavaScript, jQuery, Flask, Node.js, Express.js, React.js, Next.js, Django, Odoo ERP, REST API, FastAPI, WebSockets, Heroku, Render, BeautifulSoup

Artificial Intelligence & Machine Learning: Machine Learning, Generative AI, Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), ChromaDB, Finetuning (LoRA), Prompt Engineering, Model Evaluation, Vector Embeddings, Knowledge Base Construction, Neurosymbolic Research

Data Science & Analytics: R, RStudio, Pandas, NumPy, Scikit-learn, Data Analysis, Data Visualization, Plotly, D3.js, Tableau, Power BI, Statistical Modeling, Experiment Design, Sentiment Analysis, Text Mining

Tools & Version Control: Git, GitHub, Agile Development, CI/CD, Docker, Kubernetes, AWS (EKS, S3, RDS, IAM), Hugging Face Hub, High-Performance Computing (HPC/SLURM)

EXPERIENCE

Software Developer at Media Cybernetics, Inc Rockville, MD 02/2022– Present

- Created and customized Media Cybernetics ERP systems using Odoo, performing full-stack development
- Managed DevOps tasks such as setting up CI/CD pipelines, automating deployments, and monitoring system performance using Kubernetes.
- Developed custom modules and features within Odoo to meet specific business requirements, ensuring seamless user experiences and robust data handling.
- Conducted code reviews and maintained coding standards to ensure high-quality, maintainable code.
- Collaborated with the team to develop UX/UI guidelines, including creating a comprehensive style guide to ensure consistency across all user interfaces.
- Design and implement a Company Metrics data visualization dashboard using RestAPIs, Microsoft SQL, PowerBI, batch files to collect data from multiple agile frameworks improving weekly meeting efficiency and automating business processes.

Engineering Intern at Media Cybernetics, Inc Rockville, MD 05/2021– 12/2021

- Maintain and update legacy code to ensure compatibility between multiple legacy Capture Interface versions.
- Develop and improve Capture Interfaces features using C++, software SDK, and microscope cameras API.
- Design specialized Zephyr Squad test cycles to help deliver quality testing
- Perform quality assurance testing on capture module in Image Pro, capture interfaces, and capture devices (microscope cameras).
- Apply the principles of Agile project management to control the project workflow.

PROJECTS

Share your song website – [Live](#) / [GitHub](#)

A fully functional music blogs made specifically to share music and connect people. NodeJS, ReactJS, Semantic UI, MongoDB

- Combine efficient use of Semantic UI framework and customized CSS - JavaScript to create a minimalistic, clean, responsive, and user-friendly to all devices.
- The web-app is perfectly structured using NodeJS (Express) for enhance performance allowing handle multiple requests and asynchronous and event-driven.
- All the data are oriented using MongoDB for flexibility and full cloud-based host. The web-app is deployed via Heroku.

EDUCATION

University of Maryland, Baltimore County (UMBC) 01/2025– Present ▪ M.S. Computer Science - AI/ML/NLP Focus

University of Maryland, Baltimore County (UMBC) 08/2019– 12/2021 ▪ B.S. Computer Science - Data Science Focus

Montgomery College 08/2016– 05/2019 ▪ A.A Computer Science

CERTIFICATES

Microsoft Power BI Bootcamp - Simpliv Learning (2022) - [Certificate](#)

Skill Acquired: Power BI

- Mastered the fundamentals of Power BI, including data import, transformation, and visualization.
- Developed advanced skills in creating interactive dashboards and reports.
- Learned best practices for data modeling and DAX (Data Analysis Expressions) to enhance data analysis capabilities.

The Complete 2024 Web Development Bootcamp – Udemy (2021) - [Certificate](#)

Skill Acquired: HTML, CSS, Bootstrap, Javascript, JQuery, NodeJS, Express, API, Web Design UX/UI

- Gained comprehensive knowledge in web development technologies, including HTML5, CSS3, JavaScript, and Node.js.
- Built and deployed multiple full-stack web applications using frameworks such as React and Express.
- Enhanced understanding of front-end and back-end development, as well as database management with MongoDB.

Next.js 14 & React - The Complete Guide– Udemy (2021) - [Certificate](#)

Skill Acquired: Next.js, ReactJS, Web Design UX/UI

- Acquired in-depth knowledge of Next.js and its capabilities for server-side rendering and static site generation.
- Developed proficiency in building scalable and high-performance React applications.
- Learned advanced techniques for optimizing web applications and improving SEO.

Statistics for Data Science and Business Analysis – Udemy (2021) - [Certificate](#)

Skill Acquired: Statistics, Data Science, Business Intelligence

- Mastered key statistical concepts for data science and business analysis.
- Applied statistical methods to real-world datasets for actionable insights.

100 Days of Code: The Complete Python Pro Bootcamp – Udemy (2020) - [Certificate](#)

Skill Acquired: Python, Web Scraping, Beautiful Soup, Selenium, Flask

- Completed a comprehensive Python programming course covering basic to advanced topics.
- Developed practical skills by working on 100 coding projects, ranging from simple scripts to complex applications.
- Gained proficiency in Python libraries such as Pandas, NumPy, and Flask, as well as web development and automation.

LLM Engineering: Master AI, Large Language Models & Agents – Udemy (In-Progress)

Skill Acquired: Generative AI, RAG, LoRA and AI Agents

- Build advanced Generative AI products using cutting-edge models and frameworks.
- Implement state-of-the-art techniques such as RAG (Retrieval-Augmented Generation), QLoRA fine-tuning, and Agents.
- Deploy AI products to production with polished user interfaces and advanced capabilities.